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Original Article

Neutrophil-Lymphocyte ratio and Platelet-Lymphocyte ratio as useful predictive markers of prediabetes and diabetes mellitus

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ABSTRACT

Aims: Chronic inflammation plays a central role in the development and progression of diabetes, and in the pathogenesis of its complications. The neutrophil–lymphocyte ratio (NLR) and platelet–lymphocyte ratio (PLR) are indicators of subclinical inflammation. Mean platelet volume (MPV) is one of the platelet function indices. We investigated the association of NLR, PLR and MPV with prediabetes and type 2 diabetes mellitus and determine whether or not these are reliable markers for diagnosis.
Materials and methods: Participants were divided into four groups; 1: normal glucose tolerance, 2: impaired glucose tolerance (prediabetic), 3: newly diagnosed diabetic by OGTT. 4: group is previously diagnosed diabetic without complication. Outcomes were compared between groups.
Results: There are significant differences between all groups in NLR ($p = 0.004$) and PLR ($p = 0.021$) values. NLR values are significantly higher in prediabetic (1.60 ± 0.85), firstly diagnosed diabetic (1.58 ± 0.78) and clear diabetic (2.07 ± 0.95) groups compared to normal group (1.37 ± 0.69) (mean, SD). PLR values are significantly lower in prediabetic (90.35 ± 44.34) and firstly diagnosed diabetic (86.38 ± 45.24) groups compared to normal group (100.55 ± 48.14) but significantly higher in clear diabetic group (122.45 ± 37.43). There are no significant differences between all groups in MPV values.
Conclusion: Inflammation marker NLR significantly increases in prediabetic and diabetic patients. PLR significantly decreases in prediabetes and early stages of diabetes but increases in later stages. NLR and PLR values may be reliable predictive markers in prediabetes and diabetes mellitus.
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1. Introduction

Diabetes mellitus is a frequent chronic disease with high morbidity and mortality [1]. It is now commonly acknowledged that chronic inflammation has a significant role in the development and progression of diabetes mellitus and the pathogenesis of its complications. Several studies have reported an increase in the levels of inflammatory cytokines such as c-reactive protein (CRP), interleukin (IL)-1, IL-6 and tumour necrosis factor (TNF)-α in patients with diabetes [2,3]. Inflammatory cytokines are produced by different cell types, released into circulation and have local, central and peripheral effects on different tissue types [4]. Two of the main conditions of prediabetes are impaired levels of fasting glucose (IFG) and impaired glucose tolerance (IGT). IFG is defined as the fasting plasma glucose level ranging from 100 mg/dL (5.6 mmol/L) to 125 mg/dL (6.9 mmol/L). IGT is a condition characterized by a 2-h plasma glucose value in the oral glucose tolerance test, ranging from 140 mg/dL (7.8 mmol/L) to 199 mg/dL (11.0 mmol/L). Another prediabetic condition is glycated haemoglobin (HbA1C) that ranges from 5.7% to 6.4% [5]. Macrovascular complications are twice as common in people with IGT as in normoglycemic individuals [6]. It has been reported that patients with IGT have a high risk of developing type 2 diabetes mellitus (T2DM) but the progression of this disease can be slowed through lifestyle changes related to diet and physical activity [7]. Therefore, it is important to perform an oral glucose tolerance tests (OGTT) on individuals with a risk of diabetes to identify the presence of high levels of IGT or diagnose manifest diabetes.

Recently, neutrophil-to-lymphocyte ratio (NLR) and platelet-to-lymphocyte ratio (PLR) have emerged as new inflammatory biomarkers of T2DM [8–10].

In addition, it has been suggested that the mean platelet volume (MPV) is a marker that is associated with the number and activity of platelets. It is reported that the MPV level to be lower in high-grade inflammatory conditions such as active rheumatoid arthritis, acute attacks of familial Mediterranean fever and active chronic obstructive pulmonary diseases [11]. This study investigated and compared the NLR, PLR, MPV and certain biochemical levels of

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participants with normal tolerance (NGT), IGT, newly diagnosed diabetes and previously diagnosed manifest diabetes.

2. Materials and method

The sample of the study consisted of 110 people (66 female and 44 male) aged 18 to 80 years. Venous plasma glucose was measured 2 h after administration of 75 g glucose in 200 ml water. OGTT was performed on 76 participants, and according to their 2-h plasma glucose levels, they were divided into the following three groups; Group 1 with NGT under 140 mg/dL (7.8 mmol/L) (n = 42); Group 2 prediabetic with IGT ranging from 140 mg/dL to 199 mg/dL (n = 25), Group 3 newly diagnosed with diabetic by OGTT above 200 mg/dL (11.1 mmol/L) (n = 9). The last group, Group 4, consisted of 34 patients that had previously been diagnosed with manifest diabetes but had no complications. Diabetes mellitus and IGT were defined according to the criteria provided by the Word Health Organization Consultation [12].

The study was approved the Clinical Ethics Committee of Erzincan University (approval number: 2/09). We got written consent from the patients. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Patients with acute infections, and diabetic complications, and those that had been diagnosed with other chronic diseases were excluded from the study.

All biochemical and haematologic analyses were undertaken in the Central Biochemistry Laboratory of the Erzincan University Faculty of Medicine, Mengücek Gazi Training and Research Hospital on the same day. The complete blood count of patients was measured using a Sysmex XN-1000 automatic haematology analyser (Sysmex Corporation, Kobe, Japan). The plasma glucose concentrations, total cholesterol (TC), high-density lipoprotein (HDL) and triglyceride (TG) were analysed on a Beckman Coulter Olympus AU2700 (Beckman Coulter Inc. U.S.A.) chemistry-immune analyser. The low-density lipoprotein (LDL) value was calculated using the Friedewald equation when the fasting triglyceride concentration was <400 mg/dl. The haemoglobin A1c levels were measured using the high-performance liquid chromatography method with an Arkray Adams A1c HA 8160 analyser (Nakagyo-Ku, Kyoto, Japan). NLR and PLR were calculated for all groups. The four groups were compared in terms of their NLR, PLR, MPV and specified biochemical markers.

3. Statistical analysis

The data was analysed using the statistical package for social sciences (SPSS) (Version 17, Chicago IL, USA). Descriptive statistics (mean, standard deviation, median, minimum, maximum, number and percentile) were generated for the categorical and continuous variables. In addition, the homogeneity of variance, a prerequisite for parametric tests, was examined using Levene's test and the normality assumption was checked using the Shapiro-Wilk test. The differences between three or more groups were compared by one-way analysis of variance (ANOVA) when the assumptions of this parametric test were met and the Kruskal-Wallis test was used when the assumptions of ANOVA were not met. The significant differences between three or more groups were evaluated using the Bonferroni correction method. The chi-square test was performed to determine the relationship between two categorical variables.When no more than 20% of the cells had expected values,

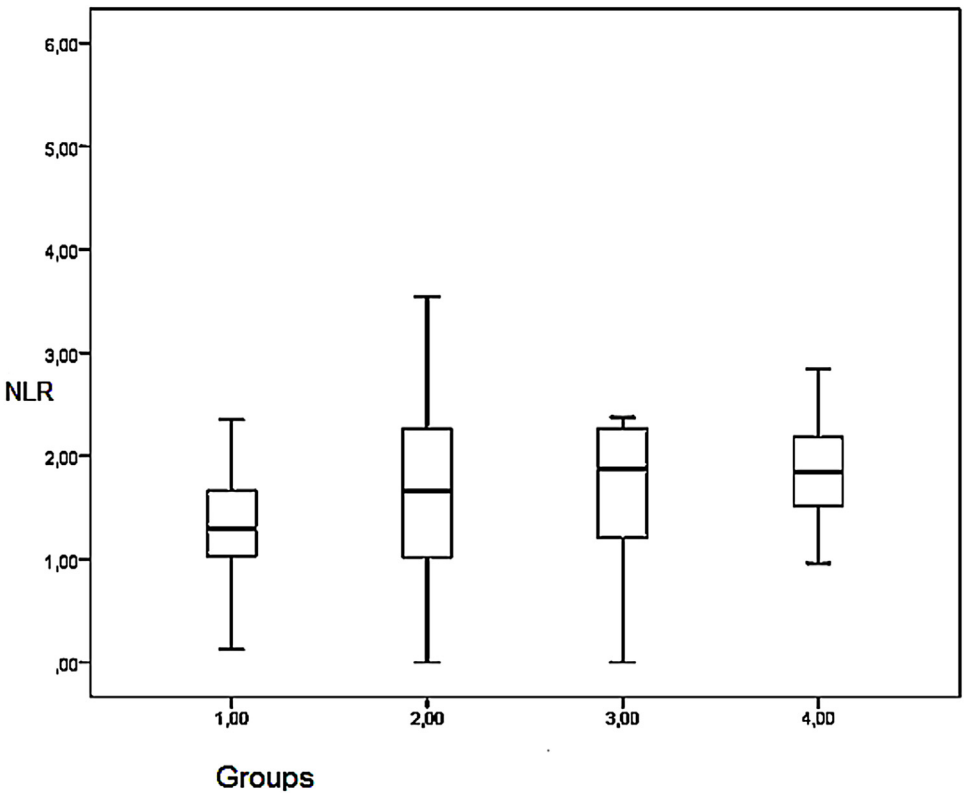


Fig. 1. Box-plot of NLR data; groups shown on the x-axis and NLR on the y-axis. Significant differences were found between all groups.

the values for these cells were calculated using the Monte Carlo simulation method. The statistical significance was accepted at $p < 0.05$.

4. Results

There was no statistically significant difference between the groups in terms of gender and age ($p = 0.317$and $p = 0.123$, respectively). The neutrophil value was lower in Group 1 than in other groups but the lymphocyte level was similar in all groups. Significant differences were found between all groups in the NLR ($p = 0.004$) and PLR ($p = 0.021$) values. NLR was significantly higher in Group 2 (IGT) (1.60 ± 0.85), Group 3 (newly diagnosed diabetic) (1.58 ± 0.78) and Group 4 (manifest diabetic) (2.07 ± 0.95) compared to Group 1 (NGT) (1.37 ± 0.69) (Fig. 1: Box-plot graphic of NLR data), and PLR was significantly lower in Group 2 (90.35 ± 44.34) and Group 3 (86.38 ± 45.24) compared to Group 1 (100.55 ± 48.14) but significantly higher in Group 4 (122.45 ± 37.43)(Fig. 2:Box-plot graphic of PLR data). There were no significant differences between all groups in terms of the MPV values.

The HbA1c value was higher in Group 4 compared to the other groups. No significant difference was observed between the groups in terms of white blood cell (WBC), haemoglobin, platelet, TC, LDL, HDL and TG. Table 1 presents the comparison of the biochemical and hematologic data between the groups.

5. Discussion

In this study, NLR was found to be higher in people with IGT, those newly diagnosed with diabetes by OGTT and those previously diagnosed with diabetes compared to individuals with NGT. Similarly, NLR was higher in people previously diagnosed with diabetes compared to those with IGT and newly diagnosed diabetics.

When the OGTT groups were compared, PLR was found to decrease most in the NGT group, followed by the IGT group, and was lowest in the newly diagnosed diabetic group. This means that PLR decreases with the increased impairment of glucose tolerance. However, PLR being found to be higher in the previously diagnosed diabetics than in other groups indicates that despite the decrease in the prediabetic and early diabetic period, PLR increases in the further stages of diabetes. To the best of our knowledge, this is the first study reporting this particular finding.

The results of the current study are in agreement with other reports that found the NLR and PLR values to be higher in the diabetic group than in the control group, and to be associated with increased and high insulin resistance[13,14]. Sefil et al. divided diabetic patients into two groups according to their HbA1C values being lower or higher than 7, and reported that increased NRL was associated with increased HbA1c [9]. Akbas et al. associated the increased NRL and PLR values in patients with diabetic nephropathy (DN) having increased albuminuria [15]. Similarly, it has been suggested that a high level of NLR is a safe predictive marker of early-stage DN [16]. In recent study, NLR has been reported to be associated with arterial stiffness and thus an increase in T2DM and diabetic retinopathy [17]. Verdoia et al. conducted a study with a large series ($n = 1377$) and reported NLR was high in diabetic patients and was related to the prevalence and severity of coronary arterial disease in diabetic cases [18]. Similarly, Aygün et al. found the prevalence of obstructive coronary artery disease to be higher in diabetic patients with $NLR > 2.05$ than those with $NLR \leq 2.05$ [19]. In another study, increased NRL due to acute myocardial infarction in diabetic patients was shown to be an independent predictor of cardiac complications such as 1-year reinfarction incidence and mortality [20]. Yılmaz et al. [21] found the NLR value to be higher in patients with morbid obesity compared to a non-obese control group. Similarly, NLR was higher in morbid obesity cases with diabetes than obese individuals without diabetes. The authors also detected a significant correlation between NLR and

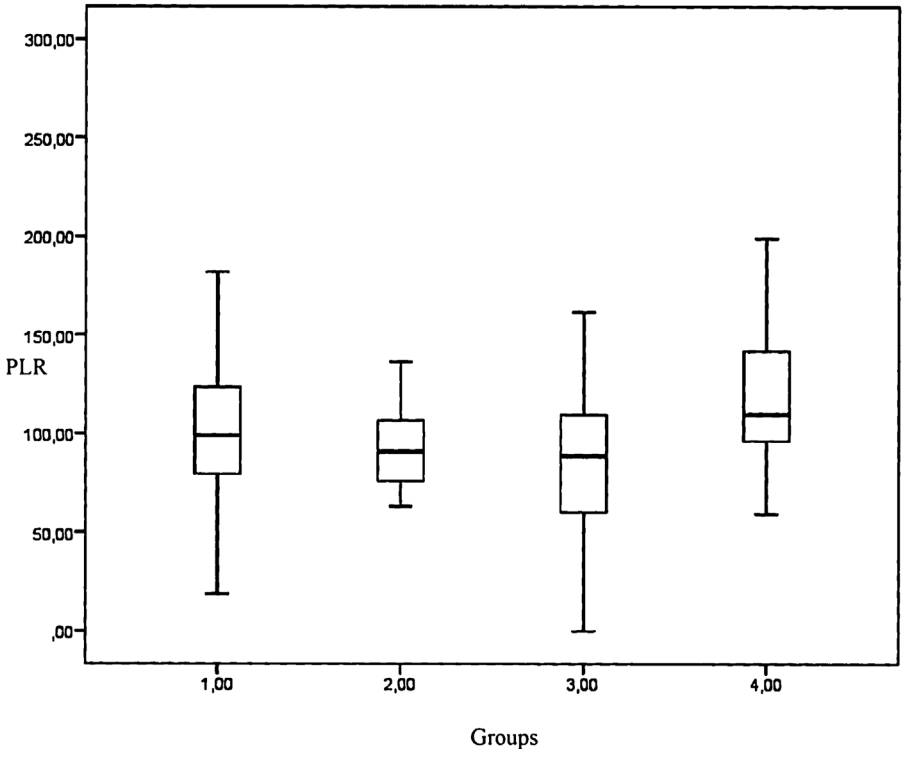


Fig. 2. Box-plot of PLR data; groups shown on the x-axis and NLR on the y-axis. Significant differences were found between all groups.

Table 1
Biochemical and hematologic data. a: Significantly different from Group 2, b: Significantly different from Group 3, c: Significantly different from Group 4; *p < 0.05, **p < 0.01.

Parameter	Group 1 (NGT) (mean ± SD)	Group 2 (IGT) (mean ± SD)	Group 3 (Firstly diagnosed diabet by OGTT) (mean ± SD)	Group 4 (Clear Diabetic) (mean ± SD)	p − Value	Multiple comparisons	
						Group	p − Value
Fasting Glucose (mg/dl)	91.88 ± 10.11 abc	103.76 ± 16.13 bc	117 ± 28.91 c	163.65 ± 97.62	0.001**	1–2 1–3 1–4 2–3 2–4 3–4	0,02* 0,01* 0,04* 0,02* 0,01* 0,04*
OGTT 2. Hour Glucose (mg/dl)	94.38 ± 29.12 ab	162.76 ± 16.7 b	253.67 ± 66.32	–	0.001**	1–2 1–3 2–3	0,029* 0,04* 0,034*
WBC (10^3 mm^3)	7.36 ± 2.66	8.08 ± 2.08	7.8 ± 3.95	8.20 ± 2.22	0.506		
Hemoglobin (g/dL)	13.13 ± 4.08	13.86 ± 4.56	12.97 ± 5.15	14.54 ± 1.9	0.390		
Platelet (10^3 mm^3)	270.24 ± 103.6	236,06 ± 107.4	231,89 ± 95.7	289,85 ± 91.1	0.142		
Neutrophil (10^3/μL)	3.65 ± 1.57	4.18 ± 2.06	4,50 ± 2.62	4,87 ± 1.81	0.044		
Lymphocyte (10^3/μL)	2.82 ± 1.91	2.52 ± 1.07	2.55 ± 1.18	2.5 ± 0.73	0.737		
MPV (fL)	9.62 ± 2.86	9.25 ± 2.97	9.26 ± 3.62	10.49 ± 0.83	0.229		
Hemoglobin A1c (%)	5.16 ± 2.19	5.26 ± 2.57	5.05 ± 3.15	7.86 ± 3.34	0.002**	1–4 2–4 3–4	0,04* 0,021* 0,001**
Total cholesterol (mg/dL)	157.88 ± 96.24	175.76 ± 96.67	126.43 ± 118.89	180 ± 79.15	0.471		
LDL (mg/dL)	97.18 ± 62.69	106.57 ± 60.7	83.57 ± 79.56	103.68 ± 50.37	0.804		
HDL (mg/dL)	38.65 ± 23.48	40.71 ± 23.93	23.86 ± 22.66	41.74 ± 18.71	0.269		
Triglycerides (mg/dL)	109.35 ± 94.58	141.14 ± 120.39	93.57 ± 100.78	171.94 ± 140.7	0.133		
NLR	1.37 ± 0.69 abc	1.6 ± 0.85 bc	1.58 ± 0.78 c	2.07 ± 0.95	0,004**	1–2 1–3 1–4 2–3 2–4 3–4	0,025* 0,049* 0,021* 0,047* 0,044* 0,034*
PLR	100.55 ± 48.14 abc	90.35 ± 44.34 bc	86.38 ± 45.24 c	122.45 ± 37.43	0,021*	1–2 1–3 1–4 2–3 2–4 3–4	0,045* 0,049* 0,047* 0,025* 0,019* 0,001**

the glucose levels at minutes 0 and 120 in OGTT. They suggested that high NLR is an independent and strong predictor of T2DM in patients with morbid obesity. These results are in agreement with the results of OGTT in the current study, which indicate that both NLR and PLR are significant predictors of NGT, IGT and diabetics.

PLR has been shown to be superior to NLR in predicting chronic infections in cases of hepatitis C infection and those with last-stage liver disease [22,23]. In the current study, we found that both NLR and PLR had a significant predictive role.

Some studies have suggested that the MPV value indicates increased platelet activity in diabetics and is correlated with insulin resistance [24,25]. In contrast, other researchers stated that there is no well-standardized test that would efficiently measure platelet activity since MPV is affected by several factors such as the time between the collection of the blood sample to the analysis, the use of anticoagulants and measurement techniques [26,27]. In the current study, there was no significant difference between the groups in terms of the MPV values.This finding is supported by the results reported in the literature indicating that the MPV value does not change in diabetic patients.

In conclusion, increased NLR and reduced PLR were found to be useful predictors of IGT and diabetes. PLR decreased in the early stages of diabetes but increased in the later stages. As in earlier studies, NLR and PLR were found to be indicators of chronic inflammation in prediabetes and diabetes.

A significant finding of this study is that the NLR and PLR values were significantly different between the OGTT groups. If this finding is confirmed by future studies with larger series and different patient groups, NLR and PLR values can be used for the diagnosis of diabetes as a simpler and cost-effective alternative to OGTT.

Conflict of interest

None.

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